

# Integrated Biodiversity Monitoring on a Seaweed Farm

|                  |                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Protocol ID      | berger_2024                                                                                                                                                                                                               |
| Sector           | Seaweed (macroalgae)                                                                                                                                                                                                      |
| Usability tier   | Tier 4 — Practitioner (farmer-runnable; some analysis partner-supported)                                                                                                                                                  |
| What it monitors | Native epibiota on farm structures and cultivated biomass; mobile fauna at the farm; farm-versus-reference community composition. Marine-mammal presence is also recorded, as supporting evidence of native mobile fauna. |
| Source paper     | Berger, C., Papazova, P., Marshall, B., Evans, F. & Williams, C. (2024). Integrating environmental and ecological monitoring with seaweed farming. bioRxiv 2024.02.15.580450.                                             |
| Associated tools | Part of a bundle: this normalized protocol, a field datasheet, and a stats workbook. (No separate calculator applies to this protocol.) Download links are added once the bundle is hosted.                               |

DOI: <https://doi.org/10.1101/2024.02.15.580450>

*Adapted into a general, gear-agnostic method for AquaPIP. Open-access CC BY-ND 4.0 preprint (not peer reviewed).*

## Introduction

Seaweed farms create suspended and seabed structures that can act as habitat for marine life. This protocol brings biodiversity monitoring into routine farm operations using three low-cost, gear-light methods that run alongside normal husbandry: a direct visual survey of the growth on farm lines, baited remote underwater video (BRUV) for mobile animals near the farm, and passive acoustic monitoring (PAM) for marine mammals.

This paper reports that these methods were chosen to be affordable and integrable into existing seaweed-farming operations, and to capture organisms across a wide range of body size and mobility. The authors also highlight that in sediment-poor, high-current sites, sediment microscopy and environmental-DNA sampling were impractical — which is why direct observation, video, and acoustics are used here instead. The aim is a repeatable baseline of what lives on and around a farm that a time-constrained operator can maintain, and that external partners can trust.

## Parameters measured

*This protocol records two kinds of measurement. Monitoring linked to a selected indicator (green) is the data that demonstrates a MEL benefit this protocol targets — it feeds a chosen framework indicator. Context monitoring (grey) captures the conditions around that data — crop growth, the physical environment, water chemistry — so the indicator readings can be interpreted and stood behind; it is recorded but is not itself tied to a selected indicator. Each parameter below is listed with its unit and a working definition.*

■ monitoring linked to a selected indicator    ■ context monitoring

### Direct visual survey — epibiota and seaweed cover

- **Epibiont taxon identity** — the organisms growing on the line/rope and on the cultivated blade, identified to the finest practical level (family or species).
- **Epibiont cover (% area)** — percent of the surveyed surface occupied by each taxon, estimated visually on (a) the line/rope and (b) one side of the blade, per sampled individual.
- **Natural seaweed-species cover on structure lines (% cover)** — percent line coverage by each settled seaweed species (contributes to community composition).
- **Biofouling cover on cultivated blades (% blade area)** — percent blade area covered by fouling groups (e.g. hydroids / “sea fur”, encrusting bryozoans, filamentous algae, and blade bleaching/discolouration).

- **Kelp length (cm)** — holdfast to blade tip.
- **Kelp maximum blade width (cm)** — widest point of the blade.
- **Wet weight per metre of cultivation line ( $\text{kg m}^{-1}$ )** — a yield/biomass measure.

### Baited remote underwater video (BRUV) — mobile fauna

- **Taxon identity** — each animal identified to the finest practical level (family or better).
- **Nmax — relative abundance (count)** — the peak number of individuals of a taxon visible in a single video frame during a deployment; summing across taxa gives Total Nmax per deployment.
- **Individual length/size (cm)** — recorded only where a calibrated stereo-camera pair is used.
- **Footage-quality class — QA** — usable vs. not, based on whether the bait at the set standoff distance and animals to family level are clearly visible.

### Passive acoustic monitoring (PAM) — marine mammals

- **Detection-positive minutes (DPM), per species (minutes)** — the number of minutes in which the detector's software classifies a valid cetacean click train, exported per hour and per day.
- **Residency time (minutes)** — how long an animal remains detectable in a pass (optional, derived).

### Environmental context — context monitoring

- **Near-surface temperature ( $^{\circ}\text{C}$ )** — logged at the culture depth (e.g.  $\sim 2$  m), e.g. every 20 minutes.
- **Seabed irradiance / PAR** — logged at culture depth and at a reference position away from the farm, for a seabed-shading comparison (relative light units or  $\mu\text{mol m}^{-2} \text{s}^{-1}$ ).
- **Current speed ( $\text{m s}^{-1}$ )** — logged near the culture depth.
- **pH, salinity (psu)** — from a near-surface grab sample, analysed promptly.
- **Dissolved nutrients — phosphate, nitrate + nitrite ( $\mu\text{M}$ )** — from a grab sample.
- **Seabed shading (% reduction)** — PAR beneath the farm expressed relative to the reference position.

## Requirements

---

- **Personnel [estimate]:** a small boat crew of [2–3]; one person able to identify local marine taxa (or to record footage/samples for later identification by a partner). PAM data analysis and stereo-video measurement may need partner support.
- **Time per visit [estimate]:** roughly [0.5–1 h] per BRUV drop (plus travel and haul); [ $\sim 1$  h] to cut and bag line sections at two positions per longline; [10–20 min] on land per sample to estimate cover; and a periodic [15–30 min] PAM service swap.

## Sampling frequency & timing

---

- **Framework guidance.** The MEL framework sets the epibiota and mobile-fauna indicators at a minimum of once per season, and the farm-versus-reference community-composition indicator at annual — sampling “at intervals sufficient to build an accurate depiction of the process over time.”
- **What the paper reports.** This paper samples on a monthly cadence across the growing season: direct line surveys and BRUV drops roughly monthly over more than a year, and PAM in multi-week deployments (about a month each) repeated across summer/autumn, winter, and spring — on the order of 21 months of coverage, with gaps around marine works.
- **Recommended to detect an effect.** Within a year, sample at least once per season, and more often (e.g. monthly) through the growing and harvest window when colonisation and fouling change fastest. Across years, run [2–3+] years — a baseline year, then repeat years — so natural interannual variability can be separated from any farm signal.

### ► Detecting a farm effect usually needs a control site

Reliably attributing a change to the farm generally requires a control / reference site — a nearby, similarly structured non-farm location surveyed the same way (preferred, though not always possible). Where a control site is not feasible, repeated before/after measures at the farm can substitute, but the timeframe needed is often unknown, and on a small farm broader ecosystem variation can swamp a detectable signal. Reference-site choices are [bracketed] for the practitioner.

## Materials

*Materials are listed per method, beside their own steps (below). The source used specific instruments; treat each item as a role to fill, not a product to buy.*

### ► Gear varies site to site — use your best judgement

The source used specific equipment (a particular light logger, a purpose-built camera on a steel frame, a specific porpoise/dolphin click-detector and its software). Availability, budgets, and site conditions differ from farm to farm. Substitute equipment you already own or can source locally, as long as it fills the same role — nothing below is mandatory as a specific product.

- **Shared across methods:** a small boat with a hauler or capstan; anchors, floating and leaded/weighted rope, buoys, and intermediate weights for isolating gear from surface motion; a GPS; datasheets on waterproof paper; labelled sample bags/containers; a camera for records.

## Methods

*Multi-method protocol — run the three surveys in parallel and repeat them on the chosen cadence. Each block keeps its own materials with its steps.*

### Method 1 — Direct visual survey of lines and cultivated biomass

↳ *Linked indicator: supports an abundance and diversity of native epibiota — match: **Exact**. Full mapping in Appendix A.*

**Materials.** boat with a hauler; a cutting blade/knife; labelled bags/containers; a measuring board or tape [for kelp length/width]; a gridded reference tray or point-count guide for estimating percent cover [define your own]; camera for records.

- **Preparation.** Fix the sampled unit — a set line length (e.g. [0.5 m] sections) at [two] positions spaced along each longline (e.g. [~90 m] apart), replicated [n = \_\_\_\_]. Fix one cover-estimation method (gridded reference or point-count) and a minimum organism size to record [define].
- **Collection.** Haul the header/structure line and a section of cultivation line aboard. Cut all growth from the defined line length at [~2 cm] from the line. Bag and label by site, line type, and position.
- **Scoring.** For each sample, select [10] individuals at random; for each, estimate the percent area cover and identity of epibionts (a) directly on the line/rope and (b) on one side of the blade. Record settled seaweed-species cover on the structure lines and blade-biofouling cover by group (hydroids, encrusting bryozoans, filamentous algae, bleaching). Optionally record kelp length, maximum blade width, and wet weight per metre as growth context.
- **Analysis.** Per sample, tally taxa present (richness), compute a diversity index (e.g. Shannon) from the cover proportions, and mean cover per taxon; summarise by site and season and compare farm vs. reference.

### Method 2 — Baited remote underwater video (BRUV) \*

↳ *Linked indicator: supports an abundance and diversity of native mobile fauna — match: **Partial**. Full mapping in Appendix A.*

**Materials.** an underwater video camera with battery/storage for the full soak [use preferred housing/camera]; a stable weighted frame or drop-frame; a bait arm holding the camera's field of view at a fixed standoff distance [e.g. ~90 cm]; a bait container and bait [e.g. ~500 g oily/crushed bait, prepared consistently]; floating + leaded rope and an anchor/surface float to isolate the frame from surface motion; GPS.

- **Preparation.** Set and keep constant a soak time (a common standard is [~60 min]), the standoff distance, and the bait type and mass. Prepare bait the same way before every drop.
- **Deployment.** Lower the frame to the seabed near the farm, in daylight, in a consistent depth band [e.g. 8–14 m], away from cliff/edge features to avoid edge effects. Isolate from surface movement with the floating/leaded rope arrangement. Record position, depth, start time, bait, and soak.
- **Scoring.** Retrieve after the fixed soak. Classify footage quality — analyse only footage where the bait at the set standoff and animals to at least family level are clearly identifiable. Score Nmax per taxon (peak count in one frame) and sum to Total Nmax. Where a calibrated stereo pair is used, measure individual lengths.
- **Analysis.** Per site, mean Total Nmax and per-taxon Nmax, taxon richness, and (with stereo) size structure; compare across sites/seasons and vs. reference.

### Method 3 — Passive acoustic monitoring (PAM) for marine mammals

↳ *Linked indicator: native mobile fauna — marine-mammal presence (supporting evidence) — match: Indirect. Full mapping in Appendix A.*

**Materials.** a self-logging cetacean click-detector [use an available static acoustic detector] with battery/storage for a multi-week deployment; its analysis software; a mooring that holds the detector clear of the seabed and isolates it from surface motion (floating + leaded rope, intermediate weight/anchor); optional acoustic release; GPS.

- **Preparation.** Program the detector; confirm battery/storage cover the planned deployment (units commonly run [4–6 months] depending on settings and temperature).
- **Deployment.** Moor the detector near the farm at a defined depth [e.g. ~9 m], oriented per the manufacturer's guidance for high-current sites (e.g. held vertical). Log deploy position, depth, datetime, and settings.
- **Maintenance.** Retrieve and swap on a schedule (e.g. monthly, or per battery/storage limit). The click data live on the detector's card, not on the field sheet.
- **Analysis.** Process with the detector's train-detection/classifier software; keep detections the classifier rates as highly likely cetacean, to limit false positives. Export detection-positive minutes (DPM) per species, per hour and per day; optionally derive residency time. Interpret as presence/relative activity, not absolute abundance; note that boat traffic can confound day/night patterns.

**Across all three methods.** Run the same surveys at the farm and at a nearby reference site and compare the assemblages — this is what addresses the farm-versus-reference community-composition benefit.

↳ *Linked indicator: farm-versus-reference community composition — match: Partial. Full mapping in Appendix A.*

## Expect these failure modes and constraints (and manage them)

---

*Each constraint is grouped under the monitoring method it affects, with how to manage it. The last group covers site and study-design issues that touch all three methods.*

### Direct visual survey — epibiota and cover

- **Observer-dependent cover, limited spatial sample** — percent cover is estimated by eye, and only a few individuals at a couple of line positions are scored. Fix the cover-estimation rule and a minimum organism size before starting, and keep the same method (ideally the same observer) across visits so results stay comparable.

### Baited remote underwater video (BRUV) — mobile fauna

- **Turbid water degrades video** — in low visibility, classify and use only clear footage; expect conservative counts.
- **Bait bias** — bait attracts generalist carnivores/scavengers and under-detects herbivores, cryptic species, and juvenile flatfish that stay off the bait — so BRUV under-represents part of the community (see Appendix B).
- **Nmax saturation / inflation** — Nmax is a peak-in-frame relative index; it can be inflated by animals that linger at the bait and underestimates where a taxon is abundant.
- **Footage / ID gaps** — restricted field of view and lighting can prevent species-level ID; combine ambiguous taxa consistently and record the rule.

### Passive acoustic monitoring (PAM) — marine mammals

- **Single-detector coverage** — one acoustic detector samples a small area; patterns can differ within ~1 km, so several detectors are needed for larger farms. Boat traffic can confound diurnal detection patterns.

### Across all methods — site & study design

- **Sediment / current constraints on method choice** — at sediment-poor, high-current sites, sediment grabs and eDNA may be impractical (as the authors found) — which is why this bundle uses observation, video, and acoustics.
- **Short timeframe / small scale** — over a short series on a small farm, natural environmental variation can swamp any farm signal — treat early results as a baseline, not proof of a farm effect.
- **Infrastructure confounds the shading context** — buoy size and line spacing drive seabed shading independently of seaweed growth; keep infrastructure notes so the shading context reading isn't misattributed to the crop.

## Data handling

---

Use a consistent sample-labelling convention: site — line type / method — position — date. Enter field records into the companion workbook; the workbook's data-entry columns mirror the datasheet exactly. Archive raw video files and detector-card exports against their deployment IDs — the field sheet holds deployment metadata, not the raw readings. Photograph representative fouling/epibiont features for a shared reference set. [Insert your own site photo] where useful.

## Methods referenced from other sources

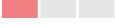
---

*Executable detail for two steps comes from cited methods rather than the source article. Resolve and verify a stable link for each before relying on it.*

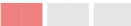
- **Stereo BRUV / Nmax for temperate, low-visibility coastal waters** — supports the video setup, soak time, and relative-abundance scoring in Method 2. Unsworth, R.K.F., Peters, J.R., McCloskey, R.M. & Hinder, S.L. (2014). *Optimising stereo baited underwater video for sampling fish and invertebrates in temperate coastal habitats*. *Estuarine, Coastal and Shelf Science*, 150, 281–287. <https://doi.org/10.1016/j.ecss.2014.03.020>
- **Acoustic click-train detection & species classification** — turns raw click data into detection-positive minutes in Method 3. Use whatever train-detection / classifier software ships with your detector. Example (the detector used here): the Chelonia F-POD app with the KERNO-F classifier. <https://www.chelonia.co.uk/f-pod/>

## Appendix A — MEL objective table

Reference: how the monitoring maps to the MEL framework. Each method links to a MEL Objective, the metric it produces, and how closely that metric matches the objective (the per-method “Linked indicator” lines above are the short form of this table). The premise: the “how” of monitoring can differ across studies as long as the metrics are the same. Framework codes appear only here.

| Goal                                                                                                                                                                                           | Monitoring method                                  | MEL Objective                                                                                                          | Metric(s)                                                                                                               | Match                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| GOAL 1 Seaweed is farmed in a way that supports an abundance and diversity of associated native species.                                                                                       | Direct visual survey of lines & cultivated biomass | Objective 1.1 (Ind. 1.1.1) — Farming supports an abundance and diversity of native epibiota                            | Abundance & diversity of epibiota — cover / individuals per species group; species & family richness; Shannon diversity | <br>Exact    |
|                                                                                                                                                                                                | Baited remote underwater video (BRUV)              | Objective 1.2 (Ind. 1.2.1) — Farming supports an abundance and diversity of native mobile fauna                        | Relative abundance of mobile fauna (Nmax) and taxon richness                                                            | <br>Partial  |
|                                                                                                                                                                                                | Passive acoustic monitoring (PAM)                  | Objective 1.2 (Ind. 1.2.1) — native mobile fauna: marine mammals                                                       | Detection-positive minutes (presence / activity) of cetaceans                                                           | <br>Indirect |
| GOAL 2 Seaweed is farmed in a way that enhances the abundance or diversity of associated native species (species associated with farmed biomass and farm site is greater than non-farm sites). | Combined biodiversity survey, farm vs. reference   | Objective 2.1 (Ind. 2.1.1) — Farming increases abundance & diversity vs. nearby, similarly structured non-farm habitat | Community-composition snapshot — taxonomic abundance & diversity + functional groups, farm vs. control                  | <br>Partial  |

**Match scale.** How closely the method's metric matches the objective's.

 **Exact** — measures the objective's metric as specified.  
 **Partial** — measures a proxy for, or subset of, the metric, or needs an added condition such as a control site.  
 **Indirect** — gives relevant evidence toward the objective (a native species present/active) but not the specified metric.

**Not mapped (context).** Kelp growth (length, blade width, yield), the physical environment (temperature, seabed irradiance/PAR, current speed), water quality (pH, salinity, phosphate, nitrate + nitrite), and seabed shading are recorded to interpret the biodiversity data but are not themselves mapped to a Habitat & Biodiversity indicator.

## Appendix B — BRUV limitations

The \* beside Method 2 refers here. Baited remote underwater video is a useful, low-impact way to sample mobile fauna, but it is biased in ways worth stating.

(1) **Nmax / MaxN is a relative index, not absolute density** — a conservative peak-in-one-frame count that avoids double-counting but saturates and underestimates where a taxon is abundant (identity-based counts have been shown to exceed Nmax by roughly 1–2.5× for some species). (2) **Bait skews the sample** toward higher-trophic-level scavengers and generalist carnivores and under-represents herbivorous, cryptic, and territorial species, and off-bait animals such as juvenile flatfish; in-water visual census typically records higher richness. (3) **There is no absolute density in moving water** — the bait plume's reach is current-dependent, so the effectively sampled area is undefined, and current speed itself affects the index. (4) **Results depend on soak time** (≈60 min is a common standard); hold it constant to compare deployments. (5) **Replicate independence**

**is not guaranteed** — mobile animals can appear on more than one nearby camera (even several-hundred-metre spacings can be insufficient). **(6) Size and biomass require a calibrated stereo pair** single-camera video also has a restricted field of view and lower clarity than an in-water observer.

**Sources:** the source paper's stated limitations, combined with current BRUV/MaxN methods literature (e.g. Dorman, Harvey & Newman 2012, PLOS ONE; Garratt et al. 2024, ICES J. Mar. Sci.; and studies comparing Nmax with identity-based counts).

## Appendix C — Provenance & practitioner decisions

---

- **Adapted from:** Berger, C., Papazova, P., Marshall, B., Evans, F. & Williams, C. (2024). Integrating environmental and ecological monitoring with seaweed farming. bioRxiv 2024.02.15.580450. <https://doi.org/10.1101/2024.02.15.580450> (CC BY-ND 4.0 preprint; not peer reviewed).
- **Indicator set:** All three indicators carried on the source row are monitored here (native epibiota; mobile fauna; farm-versus-reference community composition); none were dropped. PAM adds indirect supporting evidence for native mobile fauna.
- **Decisions left to the practitioner [bracketed]:** these are choices this protocol deliberately leaves open for the site — not gaps the original authors reported. Set them locally: sampled line length and section count; number of individuals scored per sample, the cover-estimation rule (grid / point-count), and the minimum organism size; BRUV soak time, standoff distance, bait type/mass, and depth band; whether a calibrated stereo pair is used for sizes; PAM deployment depth, mooring, and swap interval; and — critically — the choice and existence of a reference / control site. A full read of the source narrowed these but does not remove them; do not silently fill a value to look complete.